

Palamakula Deepika

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Professional Summary

AI/ML Engineer with expertise in the full data lifecycle, specializing in data preprocessing, model validation, and advanced visualization. Technically proficient in Python, AWS Cloud deployment, and IoT system development via Arduino IDE. Proven ability to transform complex datasets into actionable business intelligence using Power BI. Experienced deploying MLOps pipelines on AWS. Proven track record of increasing model accuracy by 10% and reducing manual data prep by 30% through automation.

Education

Bachelors of technology in Artificial Intelligence & Machine Learning | 2021 – 2025

Saveetha Engineering College | Chennai, TamilNadu, India | CGPA 8.2/10.0

Technical Skills

- **Programming Languages:** Python, Java, C, SQL (MySQL), Oracle SQL.
- **Machine Learning & Data:** Data Structures & Algorithms (DSA), Pandas, NumPy, Scikit-learn, XGBoost.
- **Databases & Storage:** SQL Server, NoSQL (MongoDB), Oracle Database Management.
- **Cloud & DevOps:** Google Cloud GCP, DevOps Fundamentals, GitHub, AWS (EC2, S3), Git, Docker
- **Data Visualization:** Power BI, Microsoft Excel (Advanced), Interactive Data Dashboards.
- **Design & Prototyping:** UI/UX Design, Figma, Wireframing, Power BI, Arduino IDE.

Work Experience

Machine Learning Intern | FabHost Complete Web Sol. | Jan 2022 – Mar 2022

- Developed and deployed **supervised machine learning models** using **Python** to solve complex predictive modeling challenges, achieved a **92% Precision/Recall score** using **XGBoost** and validated results with **SHAP** values., data-driven insights.
- Engineered **end-to-end data pipelines**, including **Feature Engineering** and **Model Validation**, which improved algorithmic stability and model robustness across various datasets.
- **Collaborated within cross-functional teams** to optimize internal workflows, achieving a **10% increase** in model accuracy and operational efficiency.
- Automated **data cleaning and integration processes** by developing custom Python scripts, significantly reducing manual data preparation time and streamlining the **ML Lifecycle (MLOps)**.

Projects

CareerLens AI: Scalable Career Recommendation System | Python, FastAPI, Flutter, AWS

- Architected a **3-layer AI system** (Flutter, FastAPI, Python ML) to provide personalized career path recommendations and skill gap analysis with **90%+ predictive accuracy**.
- Integrated **Explainable AI (SHAP)** with **RandomForest/XGBoost** models to offer transparent, feature-level reasoning and confidence scores for user recommendations.
- Deployed a **cloud-ready infrastructure** on **AWS EC2 and S3** using **MongoDB** for scalable data storage, ensuring modular design and seamless RESTful API communication.

Toxic Gas Leak Detection & Alerting Device | IoT, ESP32, Random Forest, IFTTT

- Developed an **IoT safety system** utilizing **ESP32 and MQ135 sensors** to monitor atmospheric toxicity and deliver real-time SMS/Email alerts via **IFTTT protocols**.
- Implemented **Random Forest Regression (RFR)** to analyze sensor telemetry and predict hazard levels, reducing potential emergency response time through automated detection.
- Engineered a **cloud-integrated monitoring solution** using **Arduino IDE and Blynk**, enabling remote device control and live data visualization of environmental metrics.

Certifications

- **Programming for Everybody (Python)** | Coursera (University of Michigan) | 2023
- **Data Structures & Algorithms (C & Java)** | Udemy | 2023
- **Fundamentals of DevOps** | Simplilearn | 2024
- **AWS Certified Cloud Practitioner** | Amazon Web Services (AWS) | 2023

Leadership & Achievements

- **Runner-Up | MSME Hackathon:** Secured 2nd place among 50+ competing teams by engineering an IoT-based Submarine Cabin Crew Alerting System designed to mitigate life-safety risks.
- **Event Organizer | Celenza 2023:** Led coordination for the flagship technical symposium at Saveetha Engineering College, managing logistics for 500+ participants.